EDUPRO

Predictive Modeling for Course Demand and Revenue Forecasting

Research Paper | Data Science & Predictive Intelligence

Abstract

EduPro requires forward-looking visibility into course demand and revenue. This study develops an end-to-end predictive modeling framework using course, instructor, and transaction information. A complete course-month panel contains 720 rows, with 480 ML-ready observations after historical lag and future-target requirements. Multiple regression models were evaluated using time-based validation. After tuning, Lasso achieved test MAE 2.8502, RMSE 3.7151, and R^2 -0.0287. The negative R^2 indicates limited explanatory power on the held-out period, so the model should be treated as decision support rather than highly accurate demand prediction. Revenue is derived transparently from predicted enrollment and course price.

1. Introduction

Online learning platforms need to anticipate demand when planning course launches, pricing, and instructor capacity. Historical reporting describes past activity but does not directly estimate future demand. EduPro therefore adds a predictive layer that combines historical course performance with course and instructor characteristics.

2. Problem Statement and Objectives

EduPro lacked predictive estimates for enrollment demand and course/category revenue. The project objectives were to integrate the available data, perform EDA, engineer leakage-safe historical features, compare regression models, tune promising models, derive revenue forecasts, and expose the results through an interactive dashboard.

3. Data and Scope

The available data contains 60 courses across 12 categories and calendar-year 2025 transaction history. The EDA summary contains 10,000 enrollments and approximately ■911,323.47 in recorded revenue. The complete course-month grid contains 720 observations; after requiring three prior months of demand history and a future target, 480 observations remain for modeling.

4. Methodology

Transactions were aggregated by course and month. Missing course-month activity was represented as zero. Features included course category, type, level, price, duration, rating, instructor experience/rating, calendar variables, enrollment lags, rolling averages, lagged revenue, and enrollment momentum. Historical features were shifted so they used only information available before the prediction month.

Models included Linear Regression, Ridge, Lasso, Random Forest, and Gradient Boosting. Categorical variables were one-hot encoded and numerical variables were imputed and standardized in reusable pipelines. A chronological train/test split and TimeSeriesSplit were used to avoid random temporal leakage.

5. Exploratory Findings

Data Science had the highest historical enrollment total at 916. Artificial Intelligence had the highest historical revenue at ■202,750.67. High-price courses generated the majority of recorded revenue, while free courses contributed enrollment without direct transaction revenue. Monthly enrollment ranged from 762 to 899.

6. Model Results

Initial evaluation showed negative R^2 across the tested models. Tuning improved the strongest candidate but did not produce strong explanatory power. The tuned Lasso result was MAE 2.8502, RMSE 3.7151, R^2 -0.0287. The mean baseline had MAE 2.8490 and RMSE 3.7130, showing that the current ML model is very close to a simple baseline.

Model	MAE	RMSE	R^2
Gradient Boosting	2.946	3.827	-0.092
Lasso	3.006	3.849	-0.104
Ridge	3.022	3.864	-0.113
Random Forest	2.958	3.872	-0.117
Linear Regression	3.028	3.872	-0.117

Tuned model	CV RMSE	Test MAE	Test RMSE	Test R^2
Lasso	3.634	2.850	3.715	-0.029
Gradient Boosting	3.657	2.866	3.723	-0.033
Random Forest	3.718	2.877	3.757	-0.052
Ridge	3.680	2.939	3.799	-0.075

7. Revenue Forecasting

Because transaction amount is linked to course price, revenue was derived as predicted enrollments multiplied by course price. Across the held-out category forecasts, actual revenue summed to approximately ■{actual_pred_rev['ActualRevenue']:,.2f} and predicted revenue to approximately ■{actual_pred_rev['PredictedRevenue']:,.2f}. This provides an interpretable business layer while preserving the limitations of the demand model.

8. Business Insights

Artificial Intelligence is the largest historical revenue category in the available summary. Data Science has the largest historical enrollment total. Free courses demonstrate why demand and revenue should be tracked separately. High-priced courses dominate recorded revenue, making price an important business dimension. Forecasts should be used alongside operational context rather than as automatic decisions.

9. Recommendations

Expand the historical time span; collect richer behavioral signals such as views, searches, conversion, refunds, and promotions; strengthen course-instructor mappings; test richer forecasting models when more history is available; use dashboard forecasts with explicit error metrics; and use scenario analysis for pricing decisions rather than interpreting observational price relationships causally.

10. Limitations

The modeling history is limited; the held-out R^2 is slightly negative; revenue forecasts inherit demand-model error; and external factors such as marketing campaigns, platform traffic, competitor pricing, academic calendars, and macroeconomic conditions are not represented.

11. Conclusion

EduPro demonstrates a reproducible predictive analytics workflow from raw transactions through demand modeling and revenue forecasting. The current data supports meaningful descriptive and business analysis, but not a claim of highly reliable next-month demand prediction. The strongest path to improvement is richer and longer historical data

combined with continued time-aware validation.